STEERING - v2.0 - SSS SSDD - SC26

Introduction

The Steering system serves as an hub for all the controls that the driver can toggle during the driving. It is fundamental as it allows to change different settings such as Controls and Power.

A first version of this system is already in use by years, but we spotted some criticism that we would like to fix in this second version.

Specifications

The steering system must include:

1. 5 push button;
2. 3 rotary switch with 10 positions (Feel free to chose whichever you like the most, they must be integrated in the PCB in a way that they don't interfere with each other);

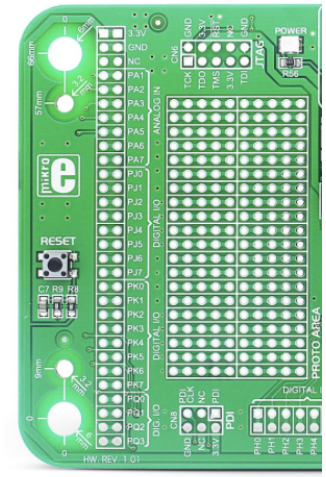
The steering system is powered by the dash via 5V.

PLEASE NOTE that the push buttons position is not already defined so you MUST place a connector for all push buttons input, so that the position of the actual button can be decided later on by not changing the PCB design;

All the signals coming from these components must be conditioned in order to work with a microcontroller (STM32F446RET6TR) that must acquire those signals and send them via CAN Bus protocol.

The dimensions of the PCB and the mounting hole position must be the same as the previous steering PCB that can be found on Altium under the name: Steering_v1_1.

Since this system has to be thought even for the driverless transition, all pins of the microcontroller must be connected to a part of the board that acts like a development board: each pin has its own soldering pad with a through hole that can be used in case of debug and in case of modularity. See the picture below as an example:



All connectors should be from molex (nanofit will work great), but if you find some better solution we can talk about it. Take into account that if it is waterproof it's a plus !

Some useful documents are listed below:

RefId	Name	Source	Description
FSG25v1.1	Formula Student Rules 2025	Rule Book	Formula student rules for the year 2025
Micro Controller	STM32F446VET6	Data-Sheet	Micro Controller Data-sheet

For any further information please refer to the last year steering system (you don't have to copy but if you don't know how to do something try to see there or in other projects) or ask Luigi. PLEASE NOTE that a lot of circuits (CAN TRANSCEIVER for example) can be copied from other projects since we already know that they work.

System External Interface Requirements

Symbol	Function	Type	Pin
CAN1_L	Can 1 Low Line	IN - IN/OUT - OUT (Chose based on what it is)	Connector X, Pin Y
GND	Ground Connection		
CAN1_H	Can 1 High Line		

Symbol	Function	Type	Pin
TV_BUTTON	Torque Vectoring Button		
TV_BUTTON_GND	Ground Connection		
TC_BUTTON	Traction Control Button		
TC_BUTTON_GND	Ground Connection		
LC_BUTTON	Launch Control Button		
LC_BUTTON_GND	Ground Connection		
GENERAL_BUTTON_1	General Button 1		
GENERAL BUTTON_1_GND	Ground Connection		
GENERAL_BUTTON_2	General Button 2		
GENERAL BUTTON_2_GND	Ground Connection		
GND	Ground Connection		
+5V	Power Supply		

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